Homework — 1.3.4 Web Technologies

Name: ______________________ Date: __________ Mark: ______ / 20

OCR H446 — set after the 1.3.4 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. The HTML below for a sports club's page contains **two** deliberate errors. Identify each error and write the correction. **[2]** *(AO2)*

```
<head>
    <title>Court Booking</titel>
</head>
<body>
    <h1>Court Booking<h1>
    <p>Book courts up to seven days ahead.</p>
    <a href="prices.html">Price list</a>
</body>
```

....

....

....

2. The club's pages contain elements like these:

```
<h1>Court Booking</h1>
<p class="note">Members may bring one guest.</p>
<p>Book courts up to seven days ahead.</p>
<div id="warning">The sports hall is closed this week.</div>
```

(a) Complete the selectors (a)–(c) in the site's stylesheet so that **every** paragraph is blue, **only** elements with the class `note` are bold, and **only** the single warning element is red. **[2]** *(AO2)*

```
(a) {
    color: blue;
}

(b) {
    font-weight: bold;
}

(c) {
    color: red;
}
```

(b) The stylesheet is a single external CSS file linked from every page. State **one** benefit of this compared with writing the same styles into the HTML of each page. **[1]** *(AO2)*

....

....

3. Number the steps from 1 to 6 to show how a search engine builds its index and answers a search. **[3]** *(AO1)*

Step	Order (1–6)
The user's search query is looked up in the index.	
The crawler follows those hyperlinks to visit and process further pages.	
Matching pages are ordered by rank (e.g. using PageRank) and the results are displayed.	
A web crawler visits a web page.	
The keywords collected are added to the index, mapped to the pages they occur on.	
The crawler records the page's keywords and metadata, and notes the hyperlinks it contains.	

4. A web form lets a user choose a username. Write **True** or **False** for each statement. If a statement is false, write a correction in the final column. **[3]** *(AO2)*

Statement	True / False	Correction if false
JavaScript running in the browser (client-side) can give the user instant feedback without waiting for a round trip to the server.		
Client-side validation code can be viewed and altered by the user, so the server must repeat any important checks.		
Checking that a username is not already taken can be done entirely client-side, without contacting the server.		

5. The **PageRank** algorithm ranks pages by importance.

(a) Explain how an incoming link from a **high-ranking** page affects a page's rank differently from a link from a **low-ranking** page. **[2]** *(AO1)*

....

....

....

(b) The rank values are recalculated repeatedly until the calculation **converges**. State what *converges* means here. **[1]** *(AO1)*

....

....

Part B — Retrieval: keep it fresh (~6 marks)

6. *(1.3.3)* State the difference between a **switch** and a **router**, referring to the type of address each uses. **[2]** *(AO1)*

7. *(1.3.1)* Explain **one** reason why **symmetric** encryption has a key distribution problem that **asymmetric** encryption avoids. **[2]** *(AO1)*

8. *(1.3.2)* A relational database contains a `Customer` table and an `Order` table. State the purpose of a **primary key**, and state how a **foreign key** links the two tables. **[2]** *(AO1)*